Controller Design

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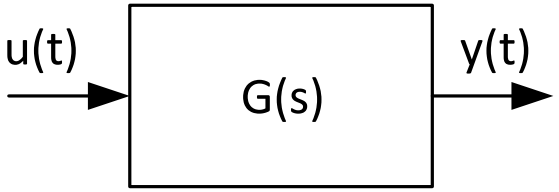
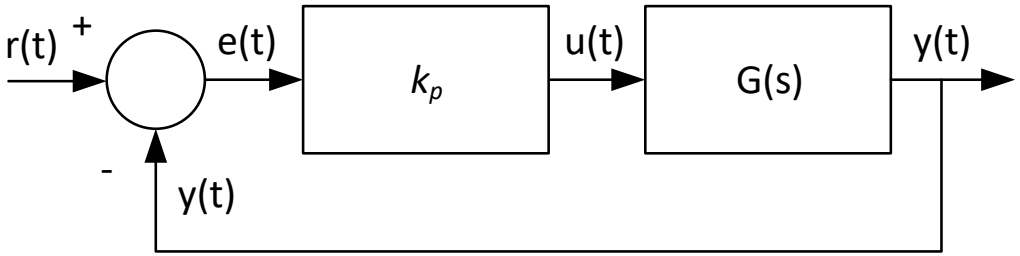
Proportional Control

- Simplest form of continuous feedback is Proportional (P) Control, where $u(t) = k_p\big(r(t) - y(t)\big)$ for some Proportional Gain $k_p > 0$.
- If the open loop plant is $G(s)$ then the unity feedback closed loop system with proportional controller in cascade with the plant is $\frac{k_p G(s)}{1+k_p G(s)}$



Open loop plant



Closed loop system with Proportional Controller in cascade with the plant

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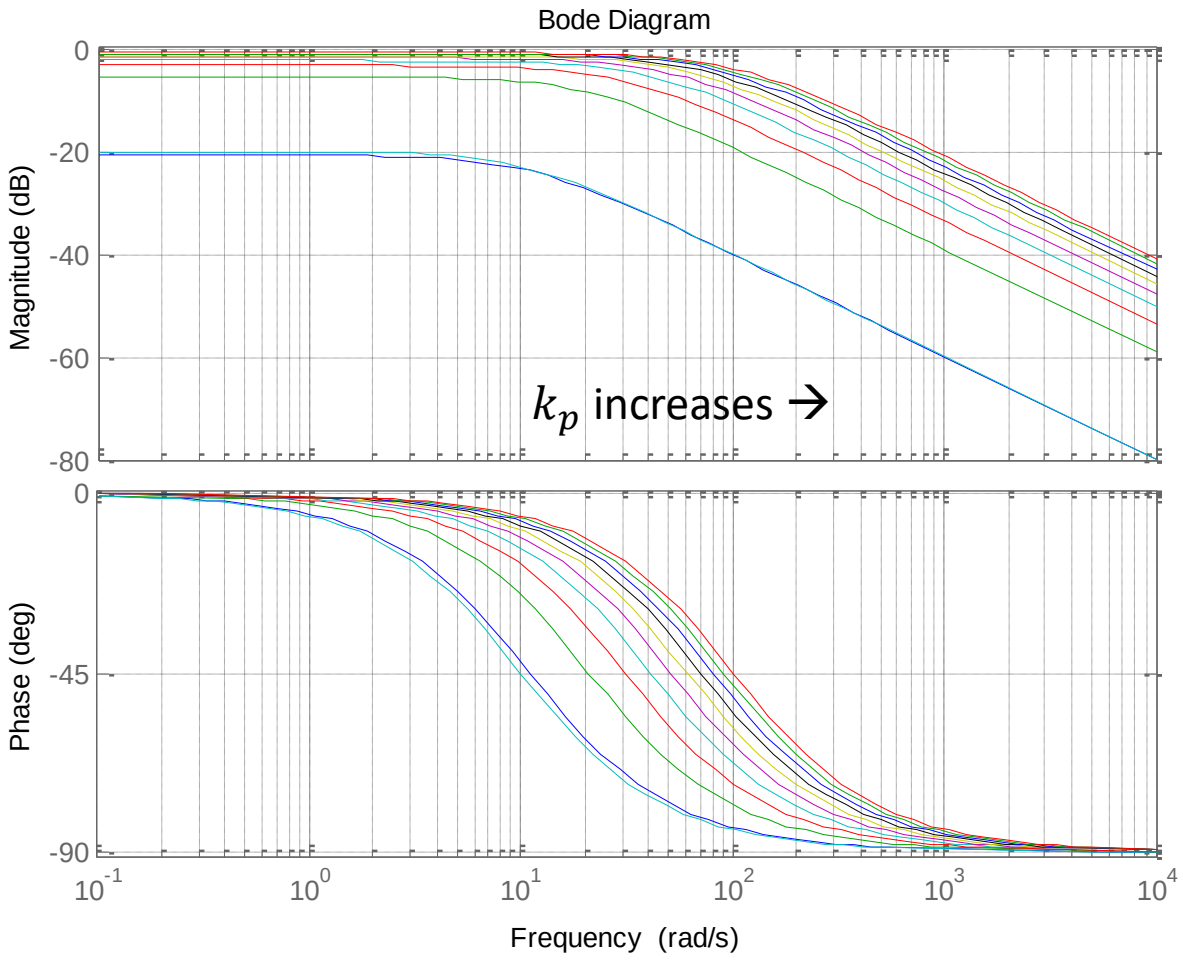
Proportional Controller with 1st Order Stable Plant

- Let $G(s) = \frac{1}{s+10}$, with $C(s) = k_p$ the unity feedback closed loop system is

$$M(s) = \frac{G(s)C(s)}{1 + G(s)C(s)}$$

$$= \frac{k_p}{s + 10 + k_p}, k_p > 0$$



$k_p = 1{:}10{:}100$

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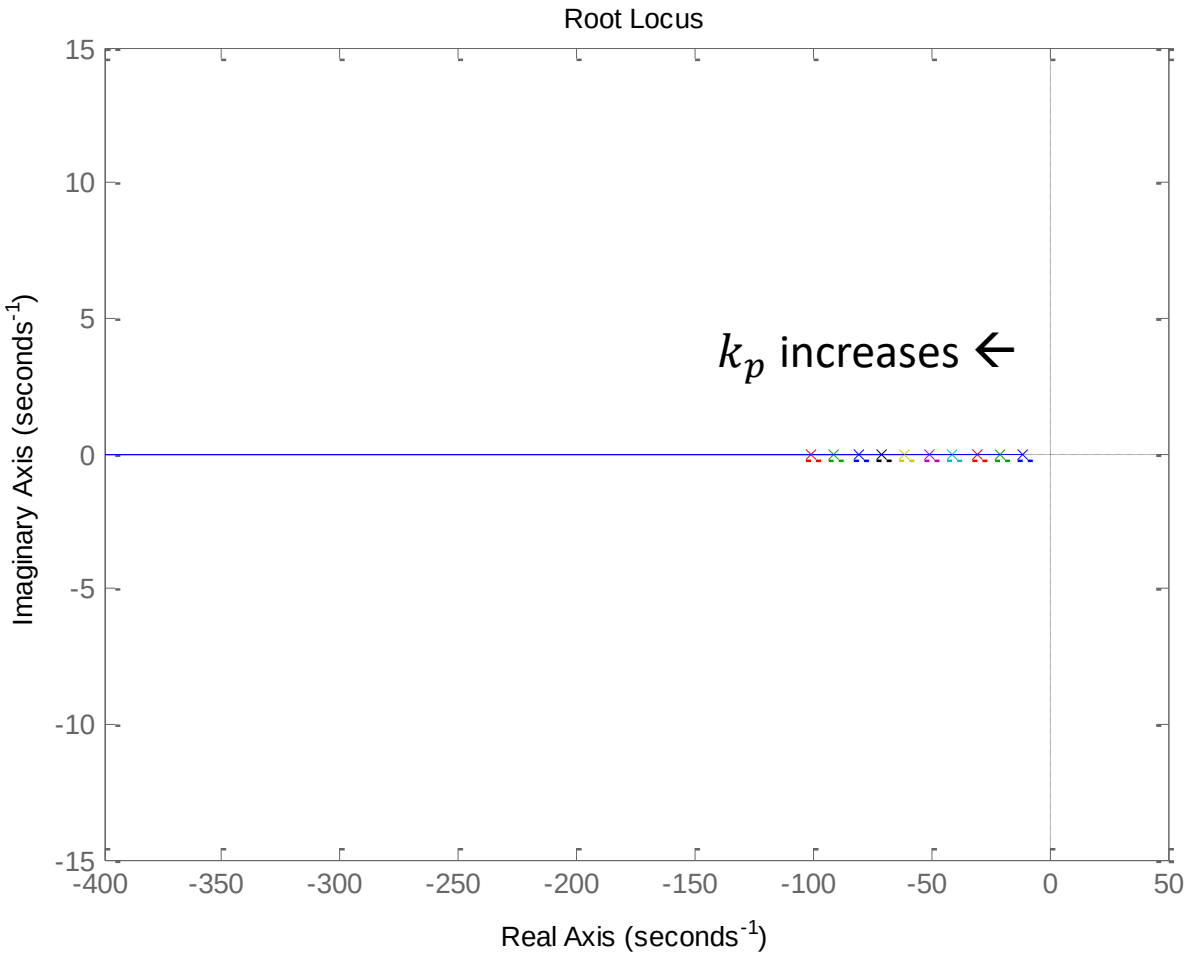
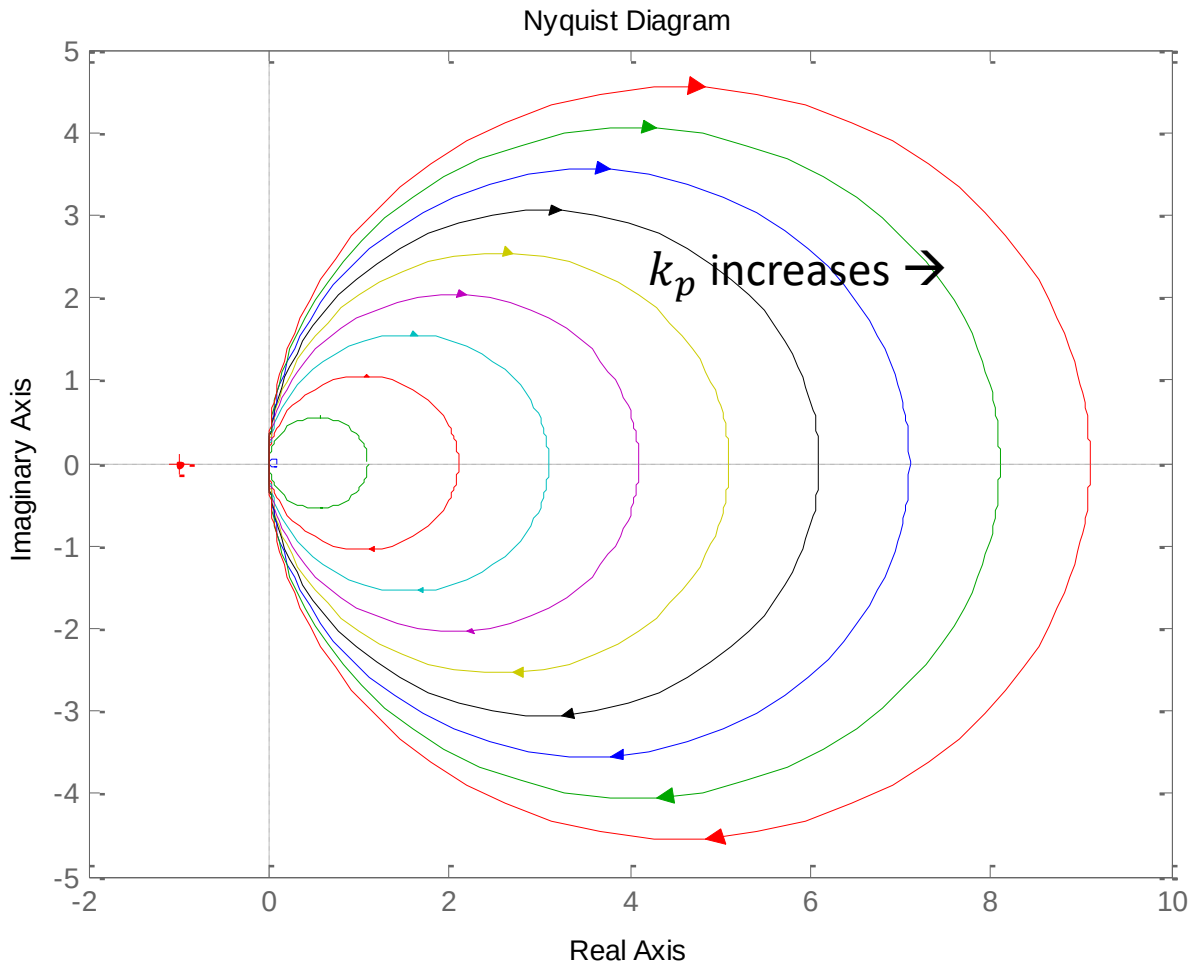
Proportional Controller with 1st Order Stable Plant contd.,





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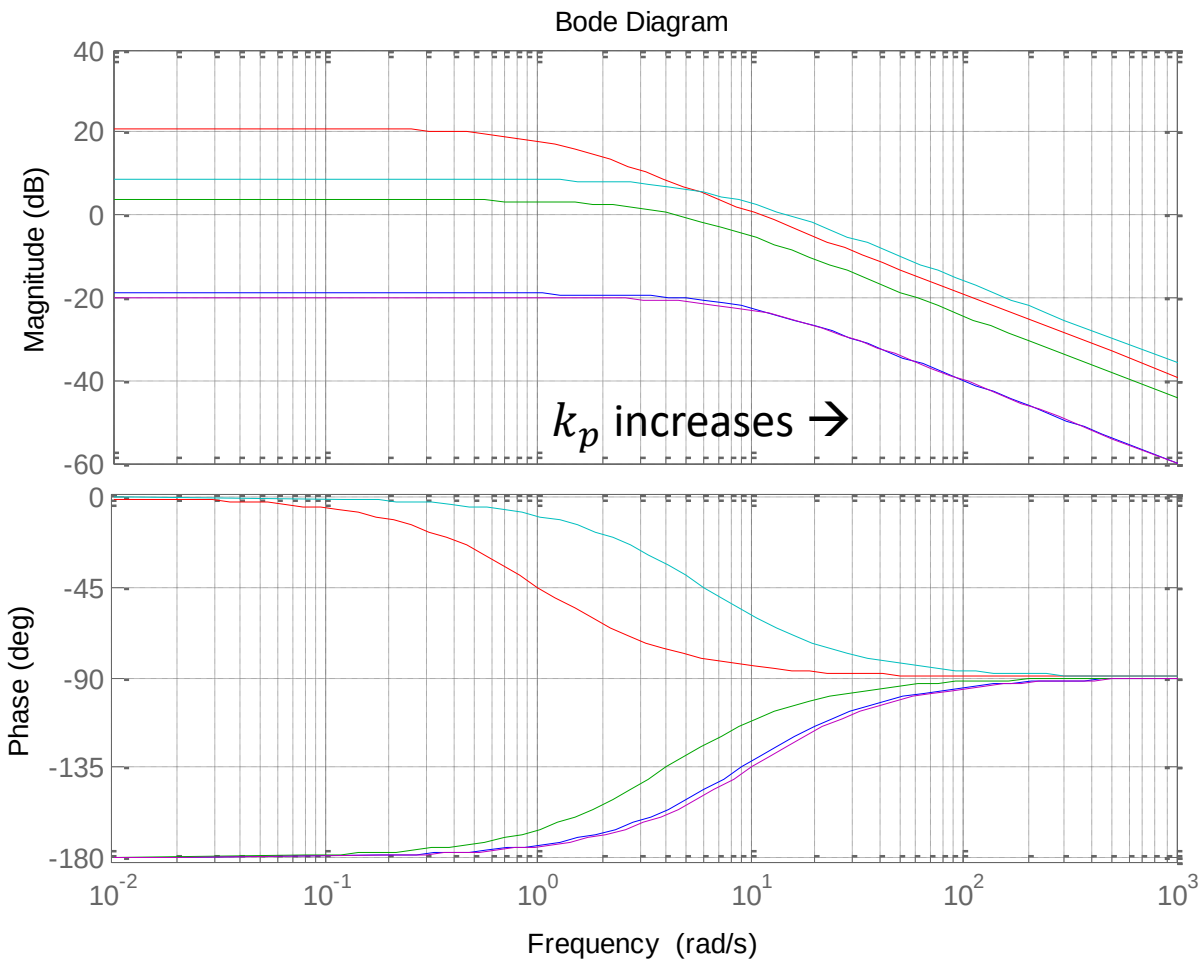
Proportional Controller with 1st Order Unstable Plant

- Let $G(s) = \frac{1}{s-10}$, with $C(s) = k_p$ the unity feedback closed loop system is

$$M(s) = \frac{G(s)C(s)}{1 + G(s)C(s)}$$

$$= \frac{k_p}{s - 10 + k_p}, k_p > 0$$



$k_p = 1:5:20$

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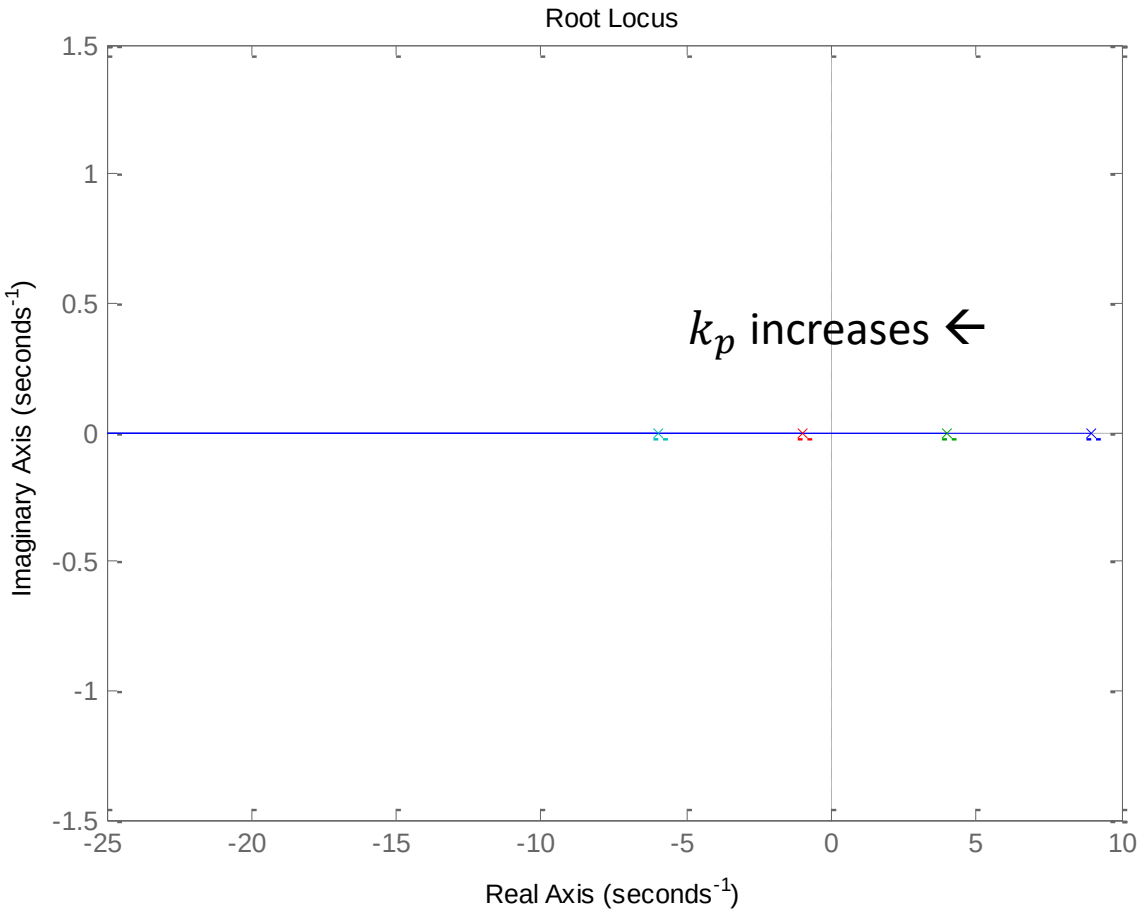
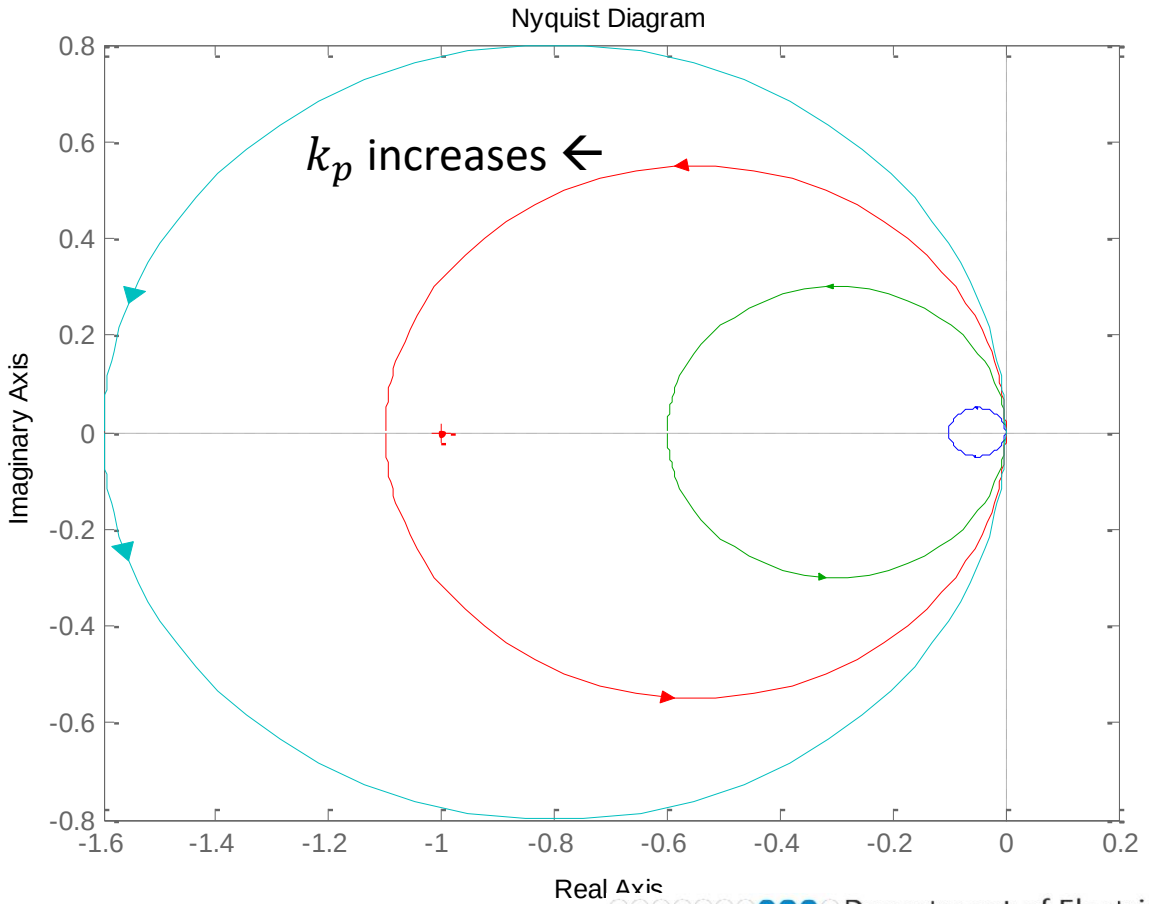
Proportional Controller with 1st Order Unstable Plant contd.,





Closed loop system unstable for $k_p < 10$

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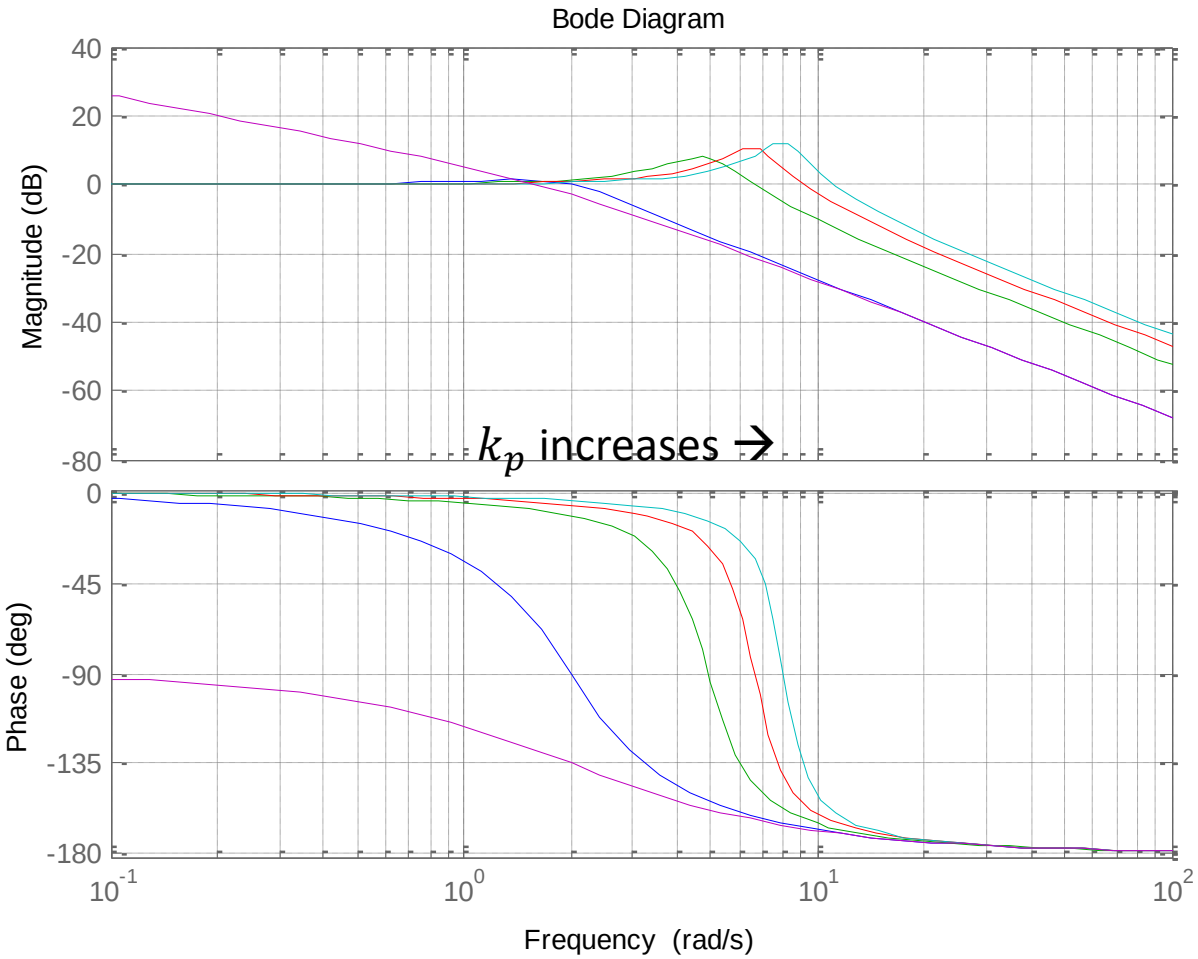
Proportional Controller with 2nd Order Type 1 Plant

- Plant $G(s) = \dfrac{\omega_n^2}{s(s+2\zeta\omega_n)}$
- Controller $C(s) = k_p$
- Unity feedback closed loop system $M(s) = \dfrac{k_p\omega_n^2}{s^2+2\zeta\omega_n s+k_p\omega_n^2}$
- $\omega_n \rightarrow \sqrt{k_p}\omega_n$ and $\zeta \rightarrow \dfrac{\zeta}{\sqrt{k_p}}$

Recall: $t_d \approx \dfrac{1+0.7\zeta}{\omega_n}$, $t_r \approx \dfrac{0.8+2.5\zeta}{\omega_n}$,

$$t_s \approx \frac{3.2}{\zeta\omega_n} \text{ for } 0 < \zeta < 0.69$$

$$t_s \approx \frac{4.5\zeta}{\omega_n} \text{ for } 0.69 < \zeta$$



$k_p = 1:5:20$

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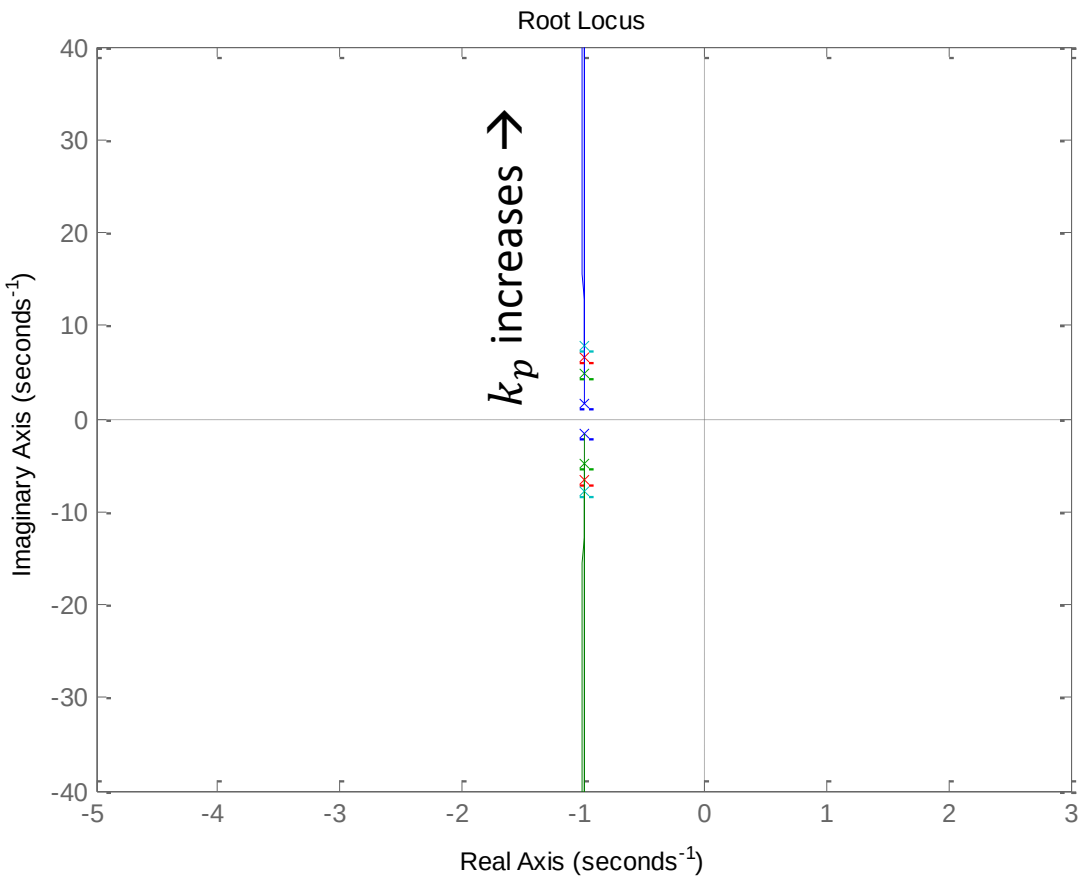
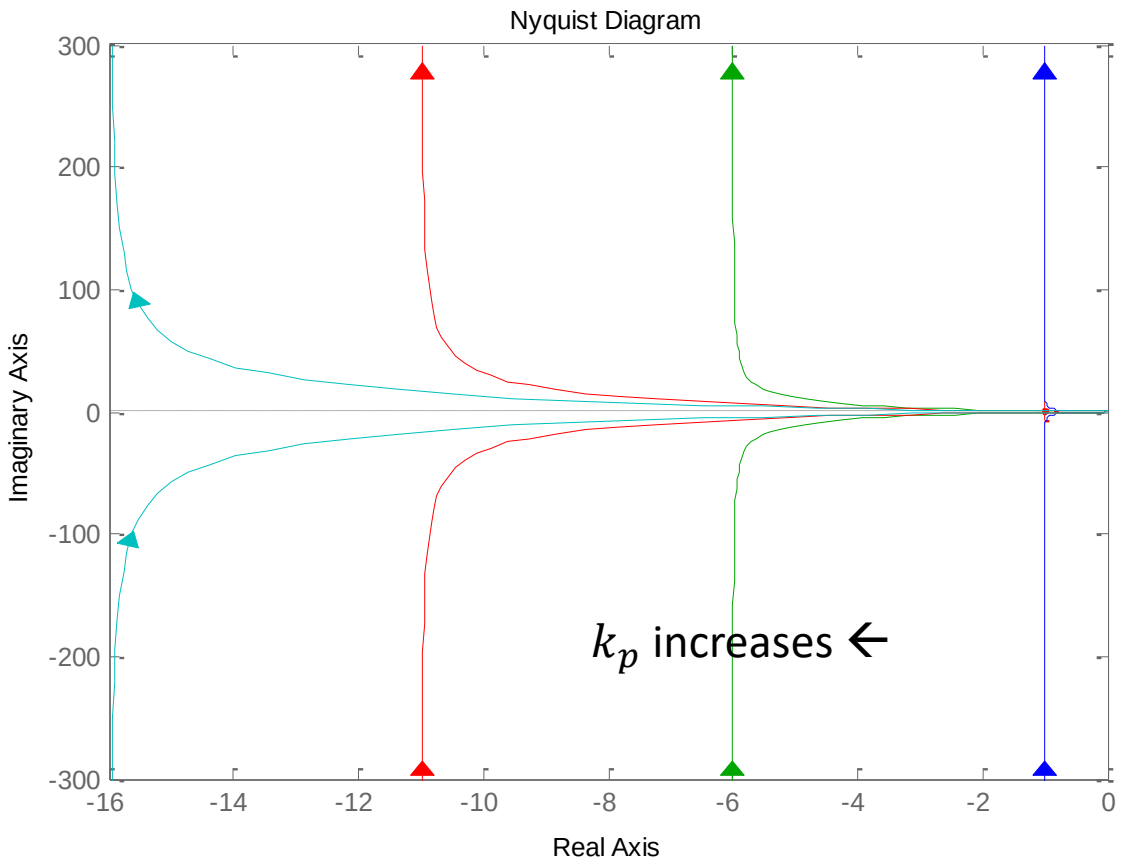
Proportional Controller with 2nd Order Type 1 Plant contd.,





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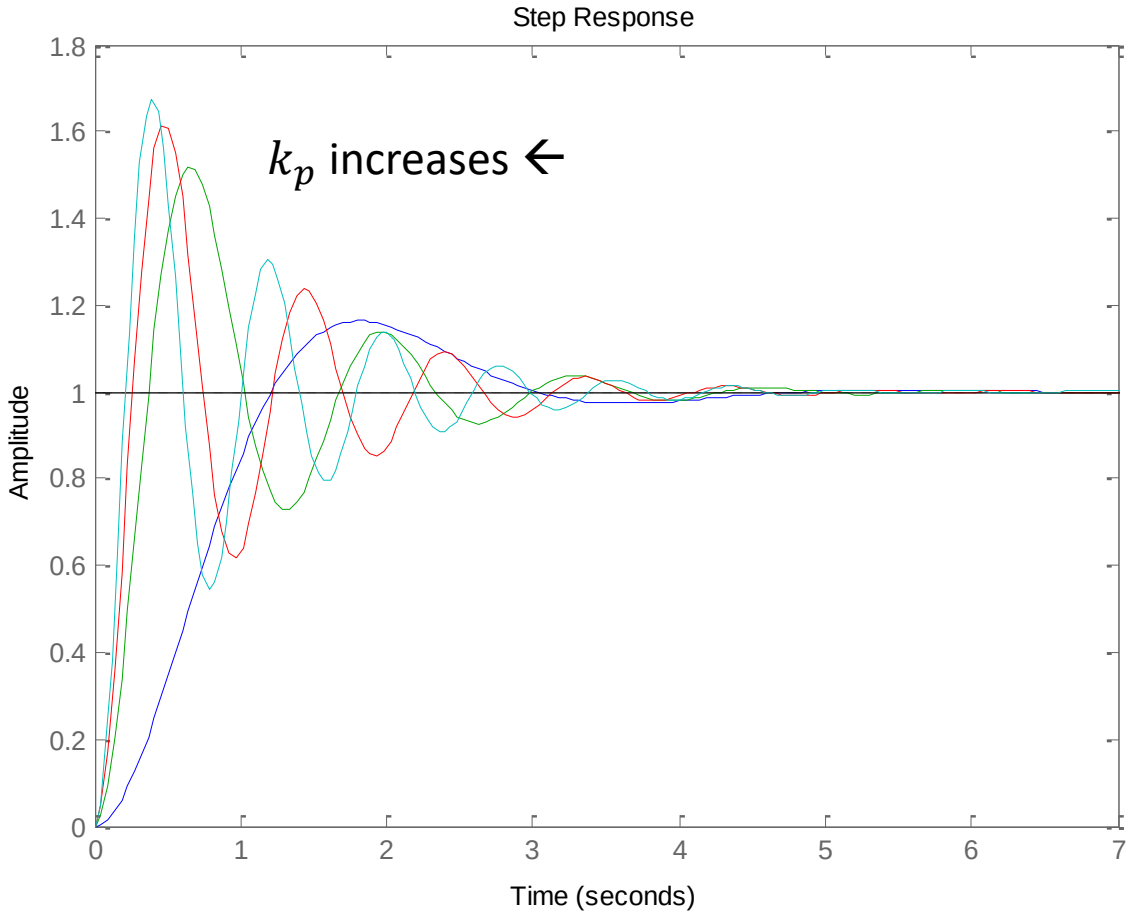
Proportional Controller with 2nd Order Type 1 Plant contd.,

- Recall: Percentage overshoot $100e^{\left(-\pi\zeta / \sqrt{1-\zeta^2}\right)}$



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Pole-Zero Cancellation Design

- Consider the same 2nd order type 1 plant $G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$
- It has poles at $s = 0, s = -2\zeta\omega_n$
- We have 2DOF if we can change ζ and ω_n independently
- If the desired quantities are ζ_d and ω_{nd} then $C(s) = \frac{\omega_{nd}^2}{\omega_n^2}\frac{(s+2\zeta\omega_n)}{(s+2\zeta_d\omega_{nd})}$ will cancel the plant pole at $s = -2\zeta\omega_n$ with its zero at $s = -2\zeta\omega_n$ and place the pole at $s = -2\zeta_d\omega_{nd}$
- This can also be seen as a Phase lead or a Phase lag controller.

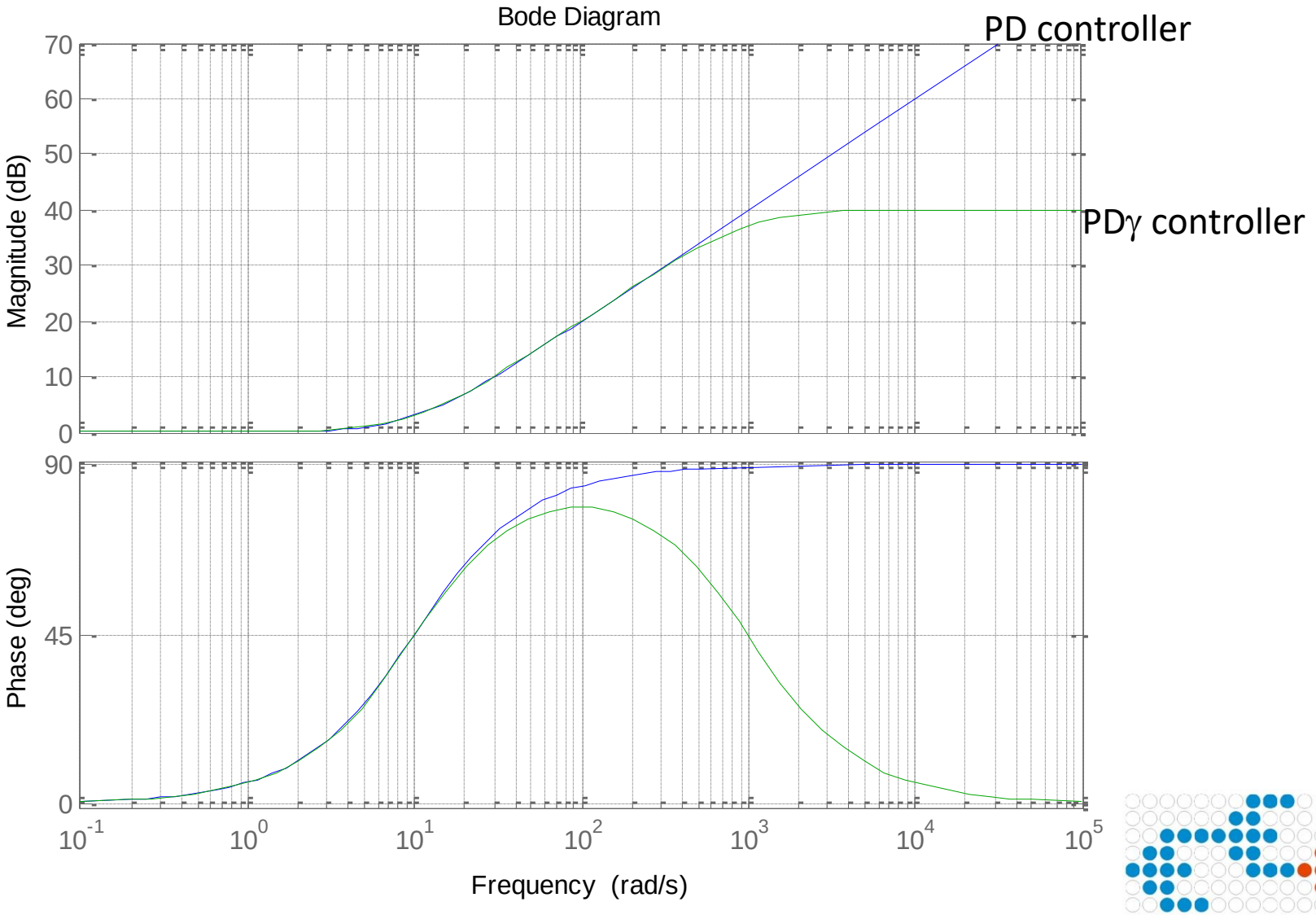
Proportional Derivative (PD) Controller

- Consider the same 2nd order type 1 plant $G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$
- It has poles at $s = 0, s = -2\zeta\omega_n$
- The pole at $s = -2\zeta\omega_n$ can be cancelled by using a zero at $s = -2\zeta\omega_n$. Hence, the controller becomes $C(s) = (s + 2\zeta\omega_n)$
- In the time domain, it is $u(t) = 1.\frac{de(t)}{dt} + 2\zeta\omega_n e(t)$. If $k_p = 2\zeta\omega_n$and $k_d = 1$, then $u(t) = k_p e(t) + k_d \frac{de(t)}{dt}$
- It contains a proportional (P) controller + derivative (D) controller. Hence it is called a PD controller
- However, PD controller is **improper**. Therefore it is used only for the design and simulation. In the implementation, it is modified to make it **proper** while not affecting the **zero location**.

Proportional Derivative Controller contd.,

- Modified version of the PD controller is called PDγ controller and it has the following structure.
- $C(s) = k_p + k_d \frac{s}{\gamma s+1}, where\ \gamma \ll 1$, hence PD controller is made proper and neither gain nor the zero location is affected.
- A pole zero cancellation design using a PD controller in the 2nd order type 1 plant $G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$ gives a 1st order unity gain closed loop system $M(s) = \frac{\omega_n^2}{s+\omega_n^2}$.
- $M(s)$ does not have an overshoot (why?), but suffers from non-zero steady state error for a step input (why?)

Frequency response of PD controller



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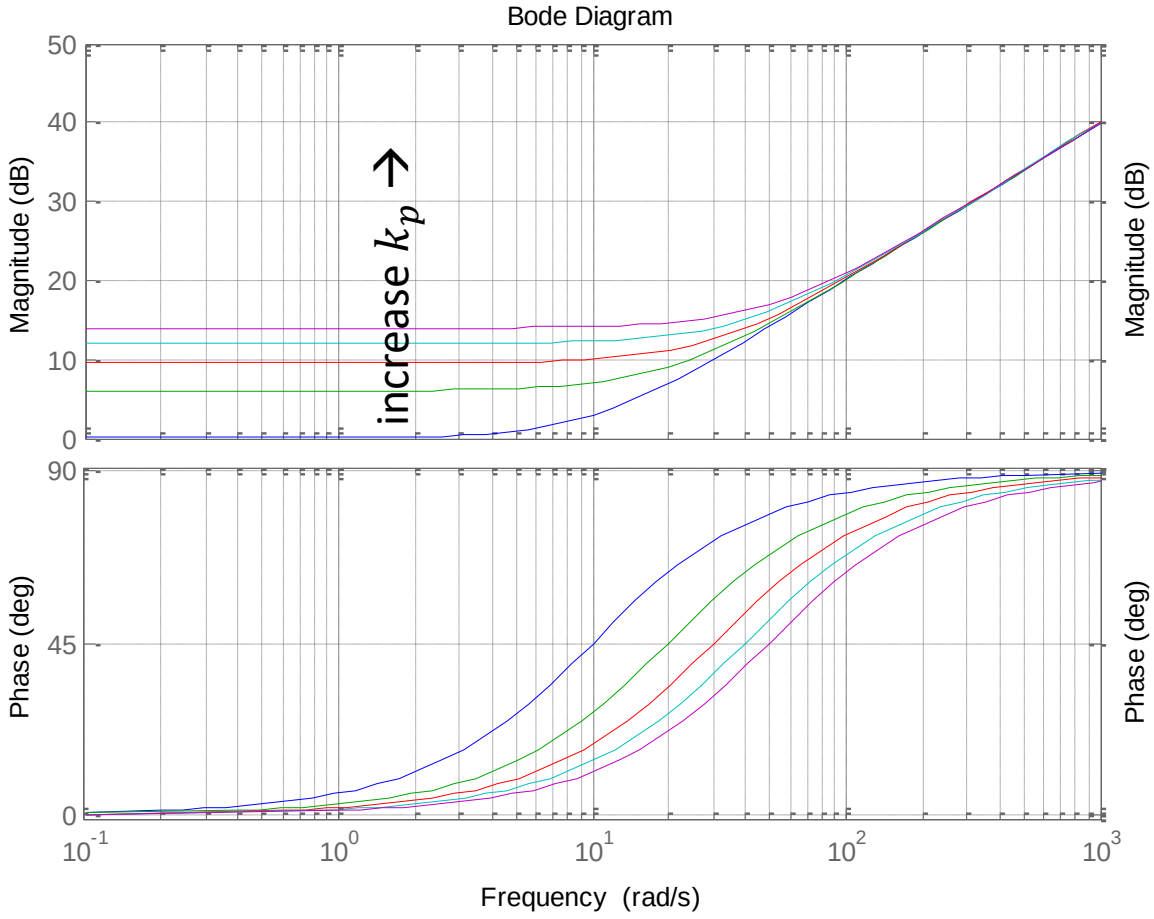
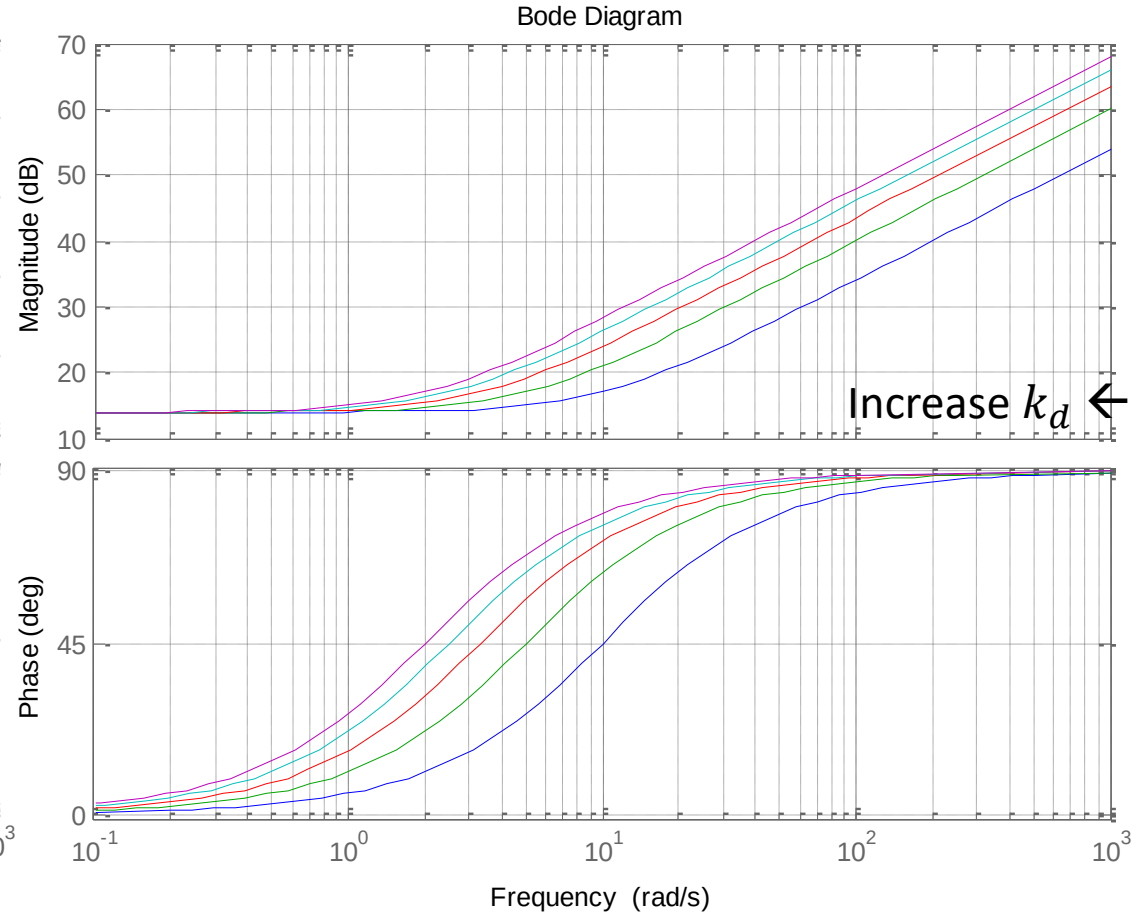
Influence of PD controller gains on frequency response

Varying k_p



Varying k_d



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Proportional Integral (PI) Controller

- Consider the same 2nd order type 0 plant $G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
- CASE I: If real, the two roots of the Characteristic Equation

$CE \equiv s^2 + 2\zeta\omega_n s+\omega_n^2 = 0$ can be expressed as $s_1, s_2 = -p_1, -p_2$

- CASE II: If complex, the two roots of the Characteristic Equation

$CE \equiv s^2 + 2\zeta\omega_n s+\omega_n^2 = 0$ can be expressed as

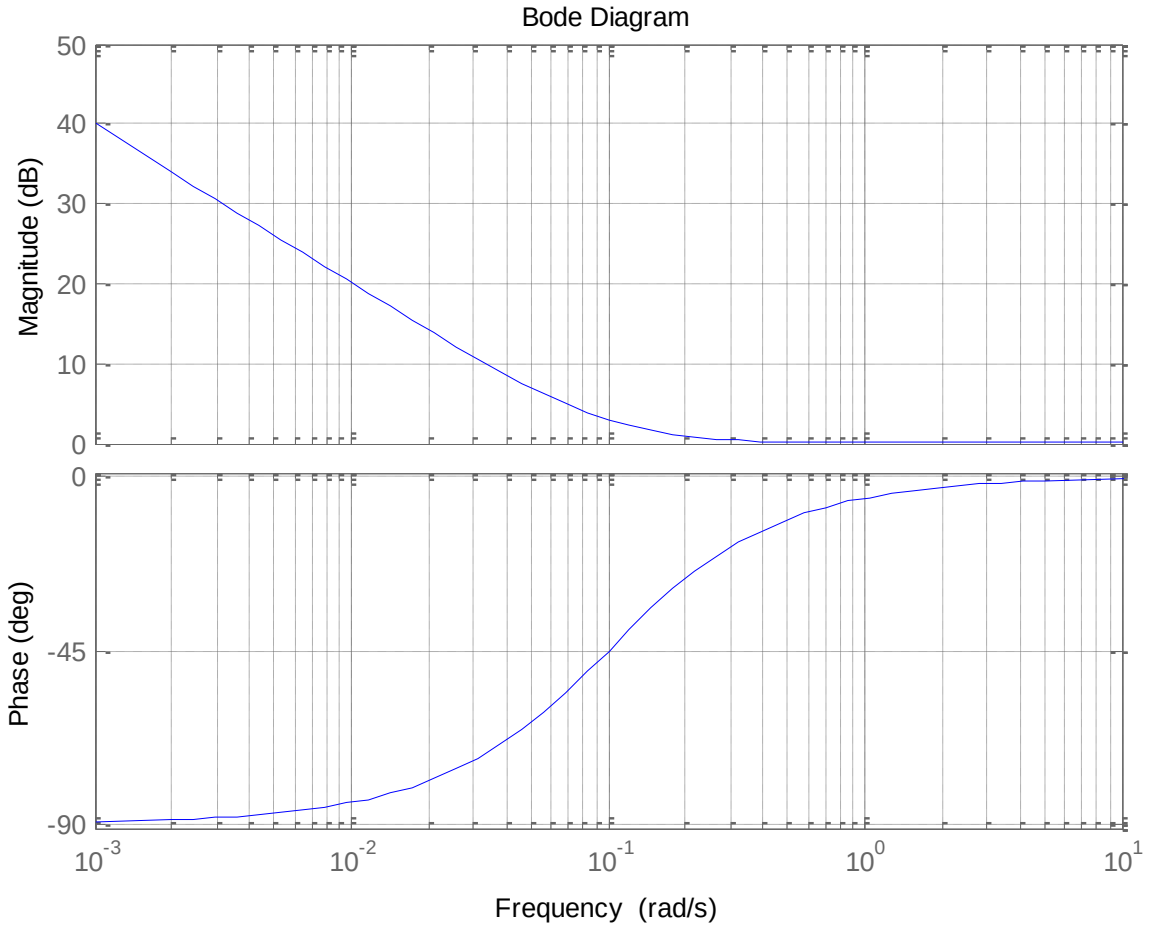
$$s_1, s_2 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\alpha \pm j\omega$$

where $\alpha = \zeta\omega_n$ and $\omega = \omega_n\sqrt{1-\zeta^2}$

- CASE I may be handled by Pole-Zero cancellation design using either a PD controller or a Phase Lead/Lag controller. However, $G(s)C(s)$ will be type 0, which will result in a non-zero steady state error for a step input.

Proportional Integral (PI) Controller contd.,

- The PI controller has the form $u(t) = k_p e(t) + k_i \int e(t)dt$, where k_p is the proportional gain and k_i is the integral gain.
- The PI controller in s-domain is $C(s) = k_p + \frac{k_i}{s}$
- Hence the PI controller usually increases the Type of the loop transfer function by 1, which ensures zero steady state error for a step input if it is Type 1



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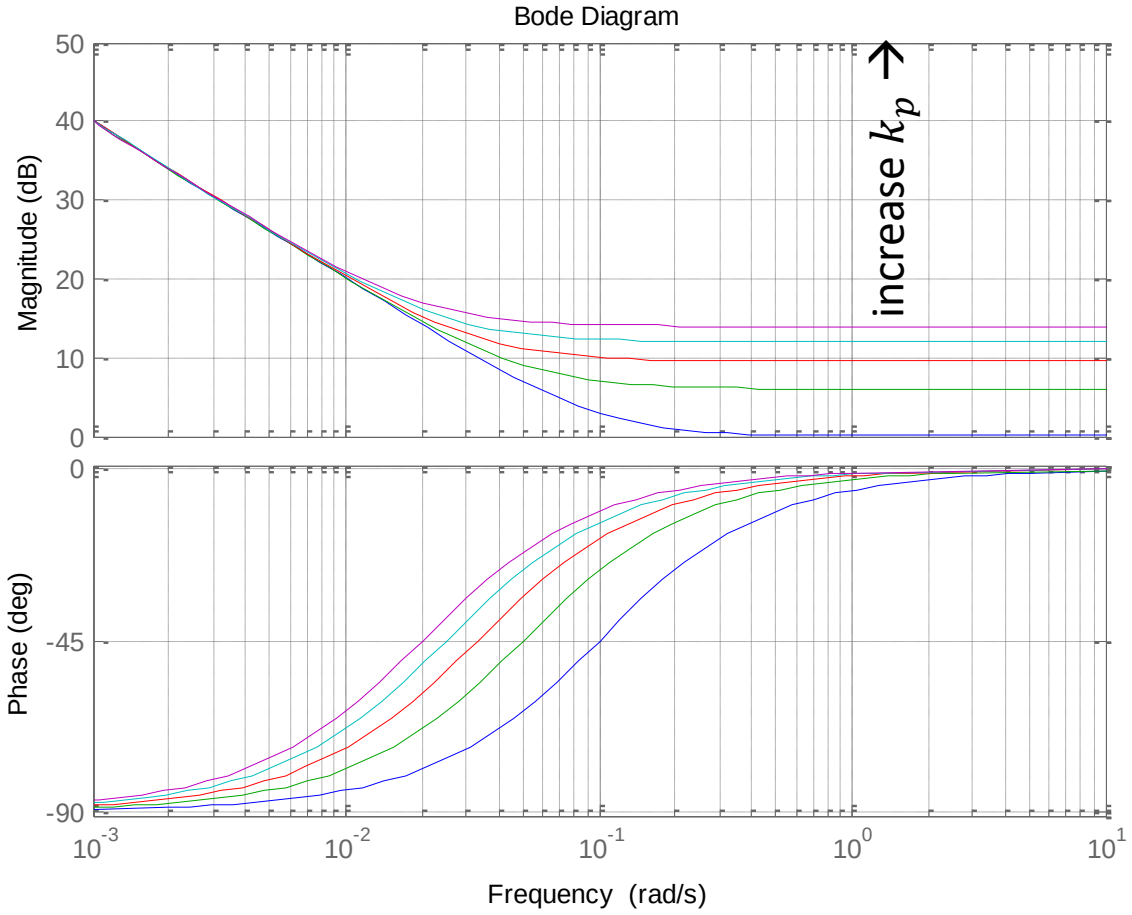
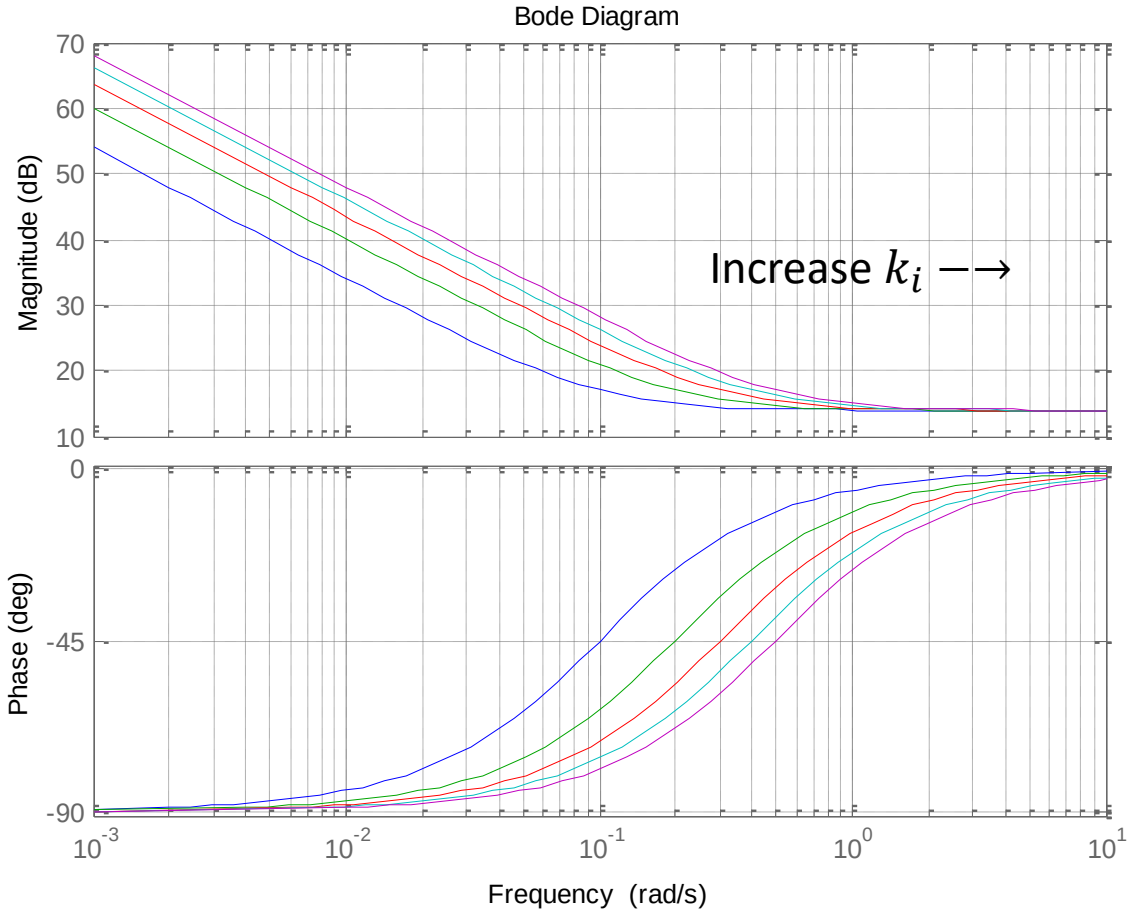
Influence of PI controller gains on frequency response

Varying k_p



Varying k_i



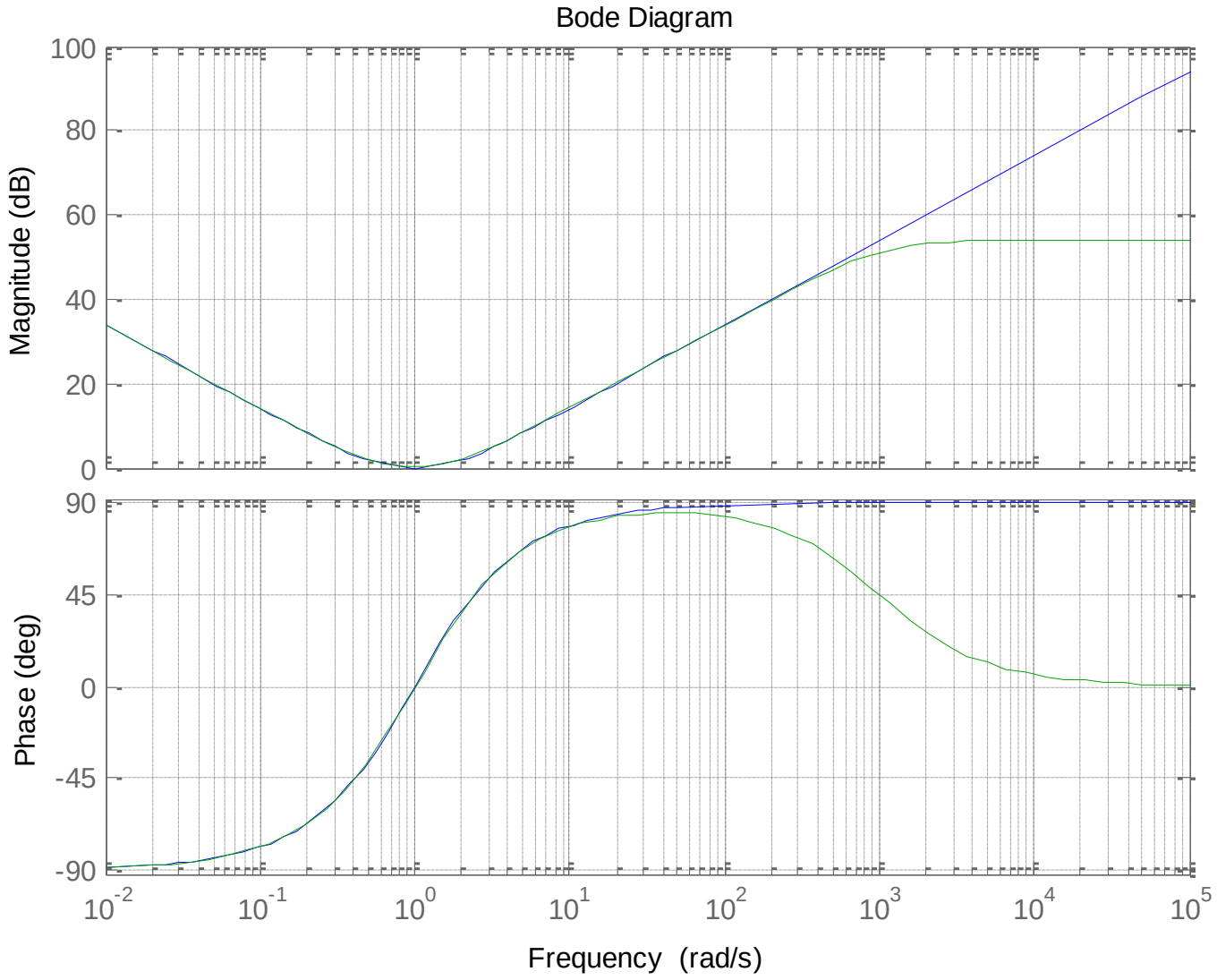
Proportional Integral Derivative (PID) Controller

- Handles CASE II above, i.e., 2nd order type 0 plant $G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$ has two complex conjugate poles
- Further, if both are real poles and they both need to be changed, PID can be used to cancel both the poles with its two zeros.
- Because of the Integrator, the loop transfer function remains Type 1, which gives zero steady state error for a step input.
- The PID controller has the form

$$u(t) = k_p e(t) + k_i \int e(t)dt + k_d \frac{de(t)}{dt},$$

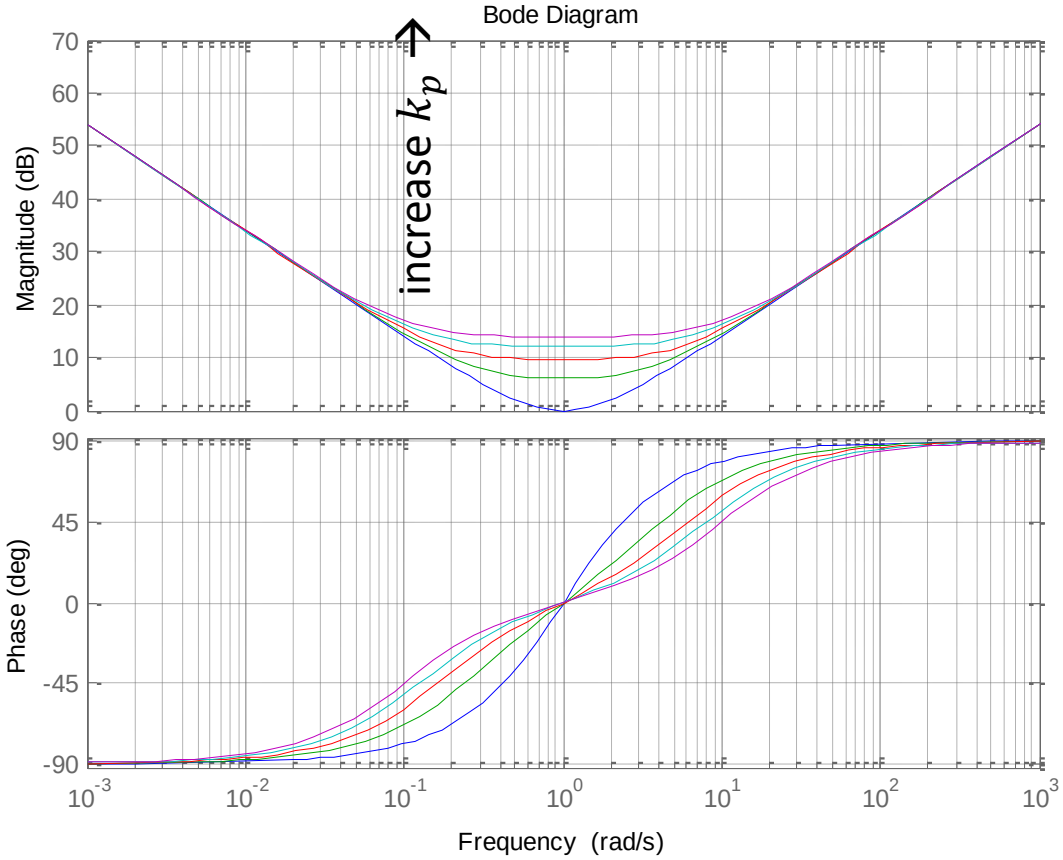
- where k_p is the proportional gain, k_i is the integral gain and k_d is the derivative gain.
- In the s - domain it is $C(s) = k_p + \frac{k_i}{s} + k_d s$
- PID controller is also improper. Like in PD, PIDγ is used to over come it.

Frequency response of PID vs. PIDγ

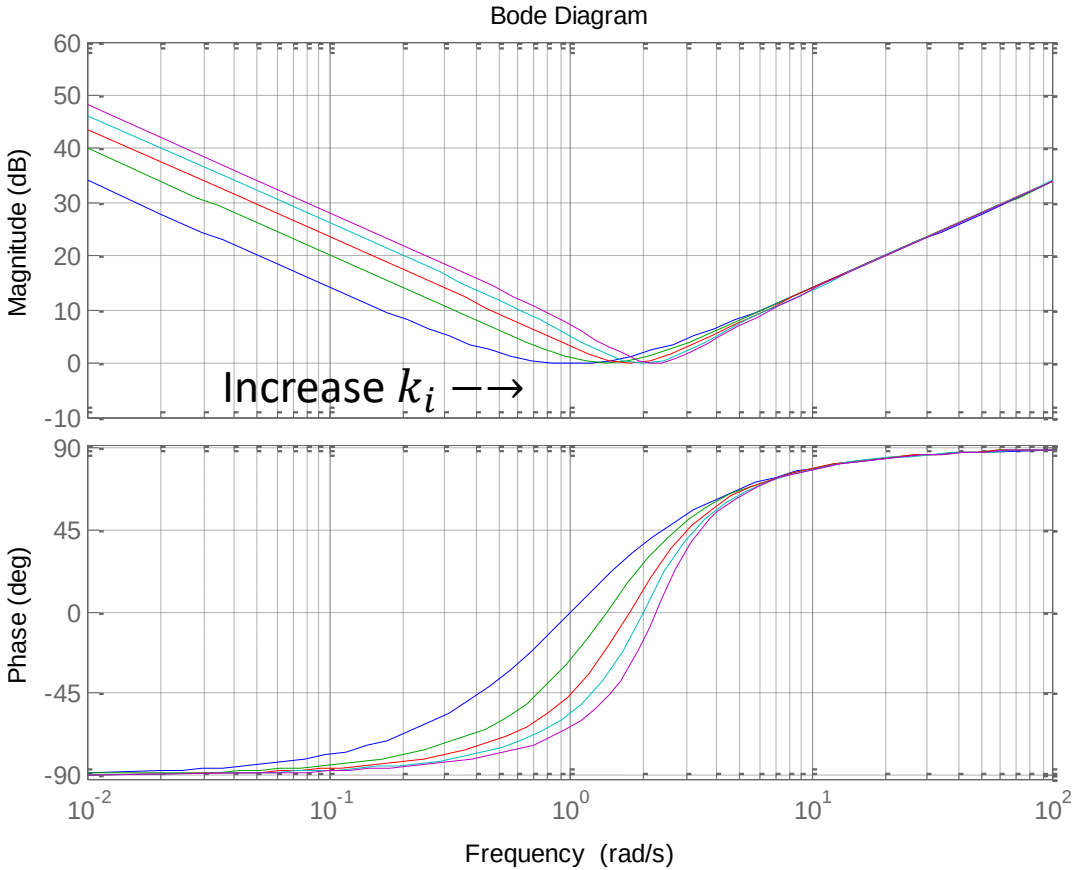


Influence of PID controller gains on frequency response

Varying Proportional Gain k_p

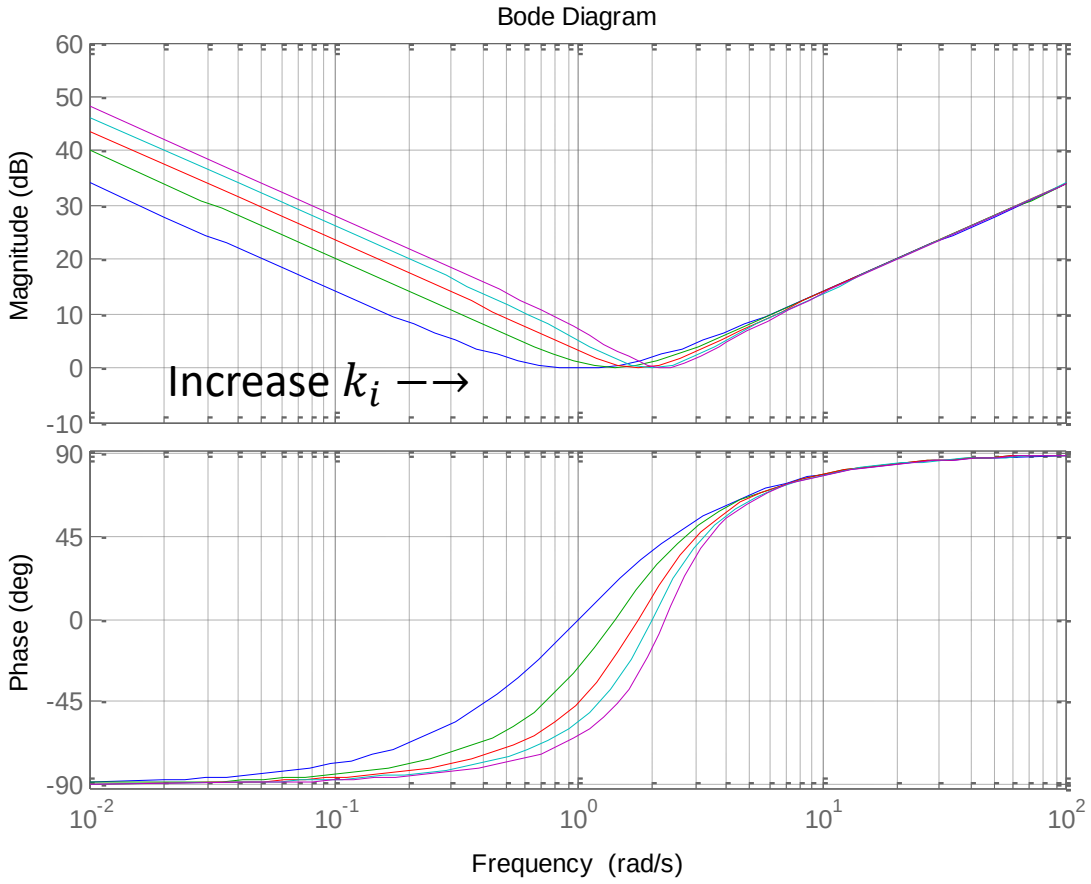


Varying Integral Gain k_i

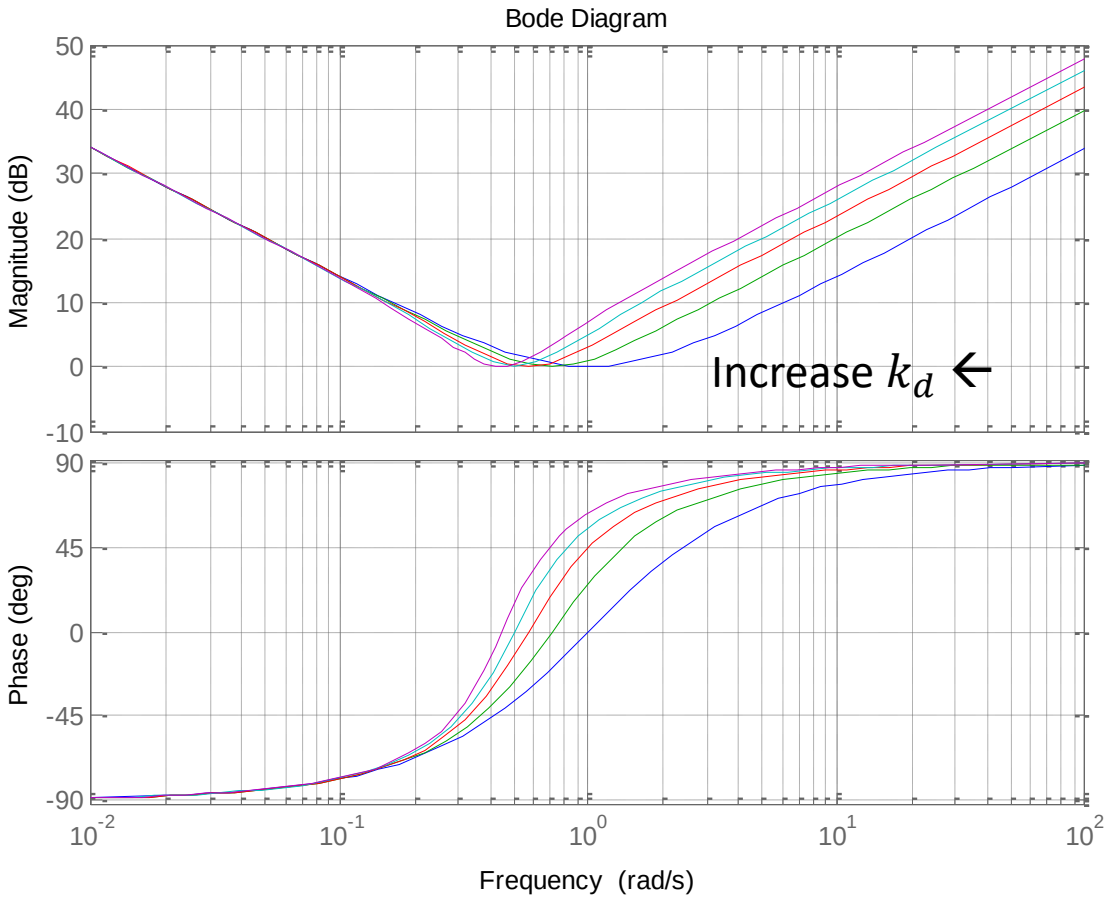


Influence of PID controller gains on frequency response cond.,

Varying Integral Gain k_i



Varying Integral Gain k_d



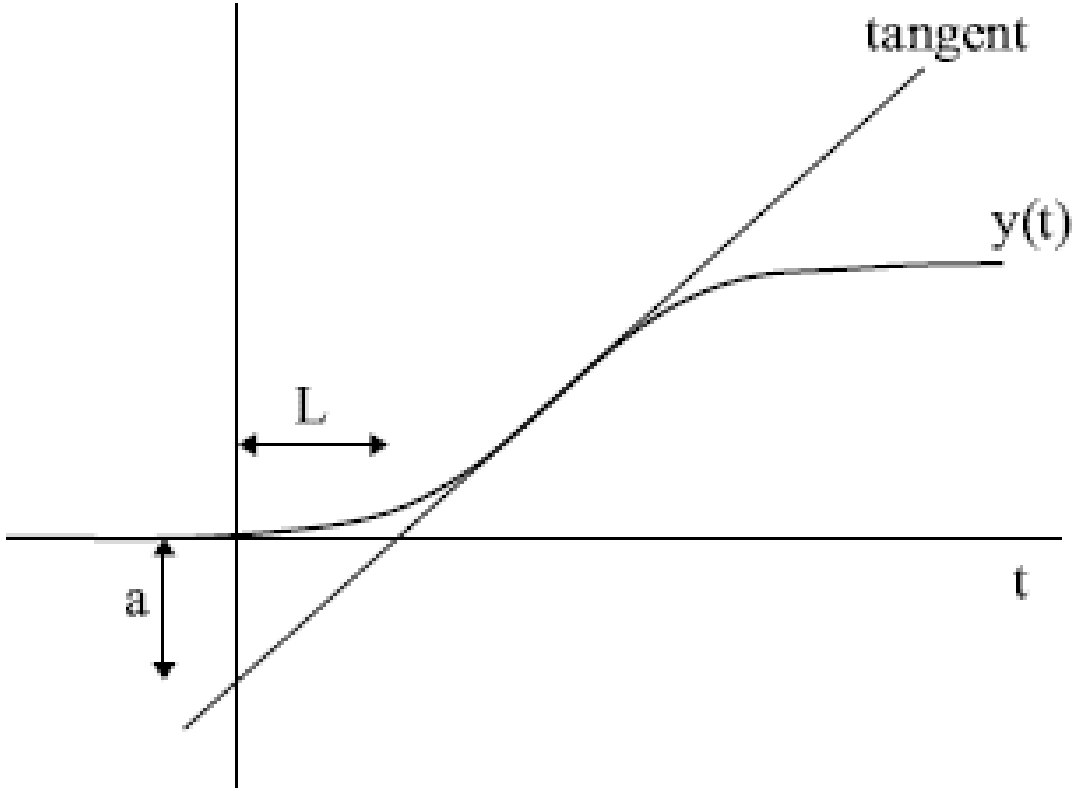
Ziegler-Nichols tuning rules

- The PID controller has the form

$$u(t) = k_p e(t) + k_i \int e(t)dt + k_d \frac{de(t)}{dt},$$

- where k_p is the proportional gain, k_i is the integral gain and k_d is the derivative gain.
- In the s - domain it is $C(s) = k_p + \frac{k_i}{s} + k_d s$
- It can also be represented as $C(s) = k_p \left(1 + \frac{1}{T_i s} + T_d s\right)$, where T_i is the integral time and T_d is the derivative time.
- Empherical rules have been set to find k_p, T_i and T_d
- They are Ziegler-Nichols tuning rules
- Two approaches: Time domain approach & Frequency domain approach

Ziegler-Nichols step response method



Controller	k_p	T_i	T_d
P	$1/a$		
PI	$0.9/a$	$3L$	
PID	$1.2/a$	$2L$	$L/2$

Ziegler-Nichols frequency response method

- Implement a Proportional Controller
- Increase k_p until the closed loop system has persistent oscillations.
- Corresponding k_p is the Ultimate Gain K_u
- Corresponding period of oscillations is the Ultimate Period T_u

Controller	k_p	T_i	T_d
P	$0.5K_u$		
PI	$0.4K_u$	$0.8T_u$	
PID	$0.6K_u$	$0.5T_u$	$0.85T_u$

Set-point tracking

- While derivative action is used to improve disturbance rejection, it creates problems in set point tracking.
- Step reference inputs may lead to sharp changes in plant input resulting in severe overshoot. This phenomenon is known as derivative kick.
- One solution is to include a pre-filter.
- Another is to allow the derivative action to act on the output only (and not the set point).
- PIDγ controller $U(s) = k_p E(s) + \frac{k_p}{T_i s} E(s) + k_p \frac{T_d s}{1+\gamma T_d s} E(s)$
- PIDγ controller treated for the derivative kick

$$U(s) = k_p E(s) + \frac{k_p}{T_i s} E(s) - k_p \frac{T_d s}{1 + \gamma T_d s} Y(s)$$

Set-point tracking contd.,

- Similarly it may be advantageous to limit the amount the proportional action acts on the set point.
- The modified controller becomes

$$U(s) = k_p \lambda R(s) - k_p Y(s) + \frac{k_p}{T_i s} E(s) - k_p \frac{T_d s}{1 + \gamma T_d s} Y(s)$$

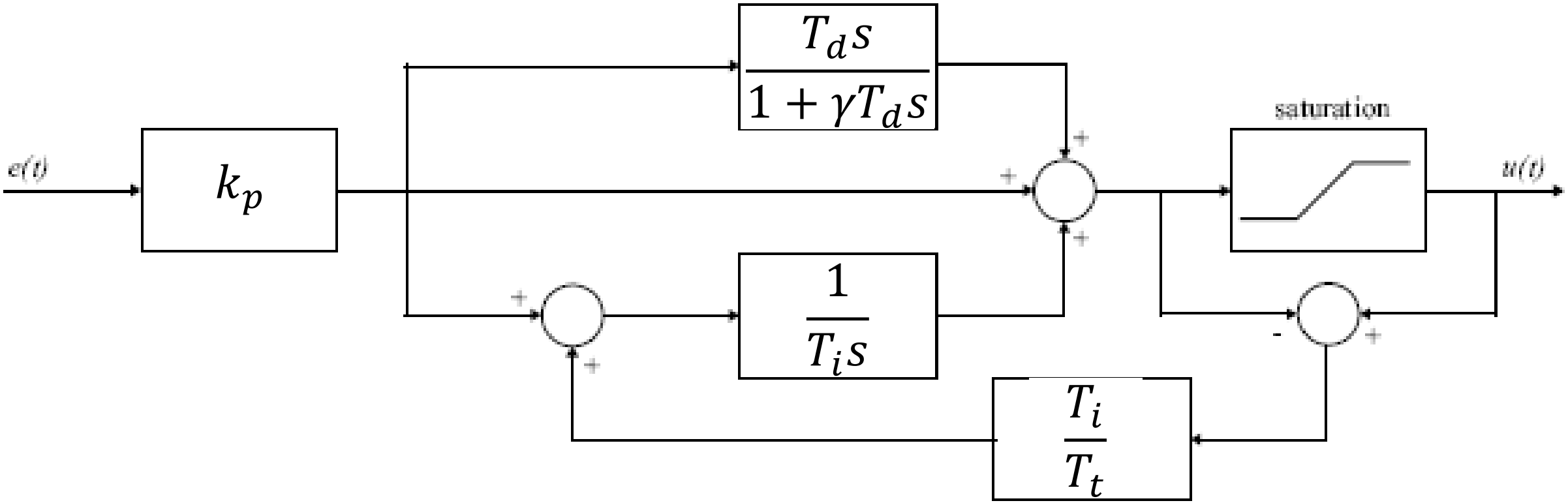
where $0 < \lambda < 1$

- Note that the closed-loop poles are not affected by these changes.
- Also the integral action continues to act on the error.

Anti-windup

- So far we have assumed that the controller out (plant input) has an infinite range.
- In practice there are always limits to actuator movement.
- For example a valve cannot be more than 100% open.
- If actuators reach their limits during normal operation, this causes degradation of performance.
- The worst case scenario is with a PI or PID controller where the set-point or disturbance is such that the actuator sits on its limit for a period of time without the output reaching its set-point.
- In this case the error will not go zero, and the I term will continue integrating. This **Integrator Windup**
- Under these circumstances it can take a long time for the control system to recover.
- To remedy this, anti-windup structures are implemented.

Typical Anti-windup scheme

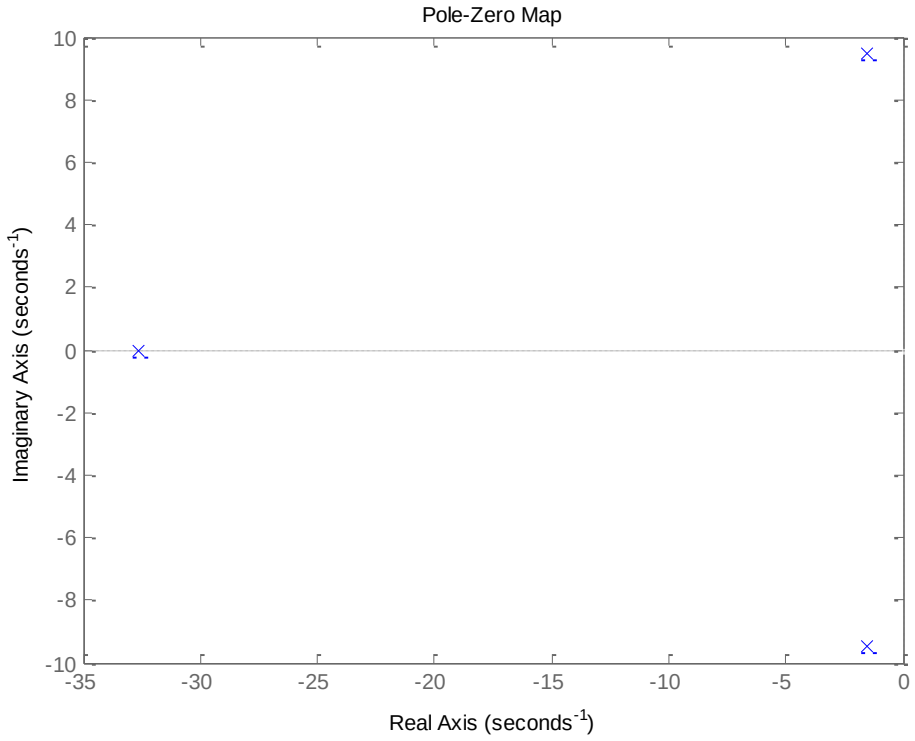


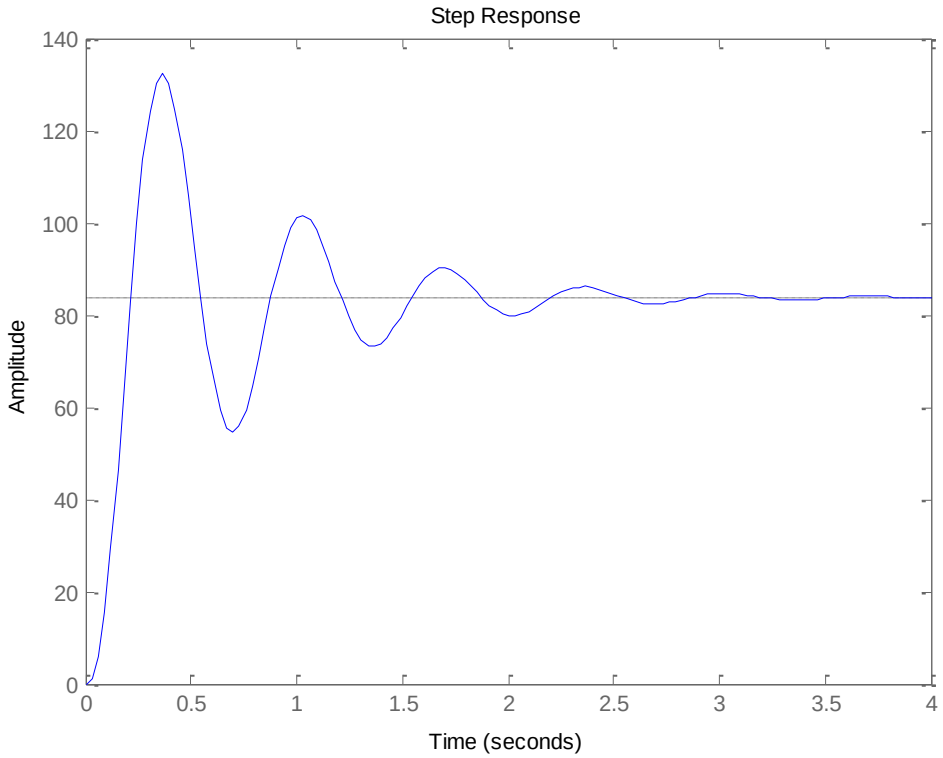
Design example 1

- A mass flow control valve has been identified to have a transfer function

$$G_p(s) = \frac{250500}{s^3+35.66s^2+190.6s+2991} = \frac{250500}{(s+32.63)(s^2+3.03s+91.67)}$$

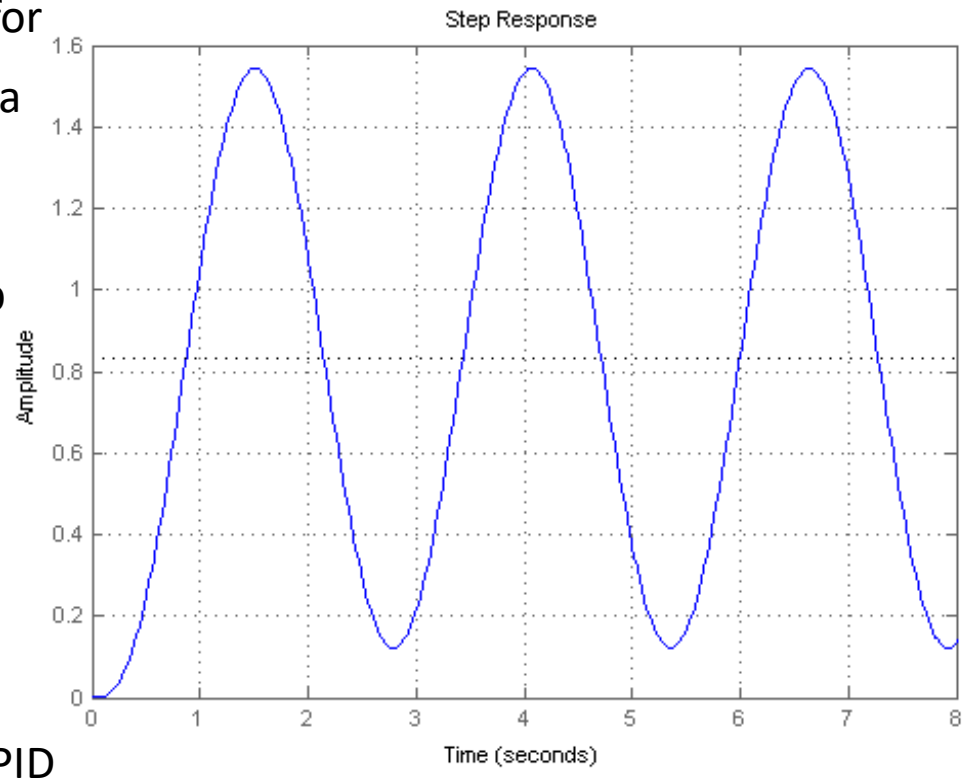
- Design a controller that ensures zero steady state error with minimum overshoot and rise time less than 0.1 s.





Design example 2

- It is necessary to design a PID controller $C(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right)$ for the system $G(s) = \frac{1}{(s+2)(s^2+2s+2)}$, where the closed loop system has a unity feedback.
- In order to find the controller parameters K_p, T_i and T_d using Ziegler Nichols tuning method, an experiment is carried out using a unit step input.
- In the experiment, with $T_i = \infty$ and $T_d = 0$, K_p is increased until the closed loop system has sustained oscillations. The corresponding output for K_p=20 is shown in the figure.



(a) Find the ultimate gain of the system K_u
(b) Find the ultimate period T_u of the sustained oscillations
(c) Use the answers in problems (a) and (b) above to find the PID controller parameters using Ziegler Nichols tuning method.

(a) K_p corresponding to sustained oscillations is the ultimate gain K_u. Hence $K_u = 20$.
(b) From the graph, the period of sustained oscillations = T_u = 2.5 s
(c) According to the Ziegler Nichols tuning rules

$$K_p = 0.6K_u, T_i = 0.5T_u, T_d = 0.125T_u$$

Hence

$$K_p = 12, T_i = 1.25, T_d = 0.3125$$